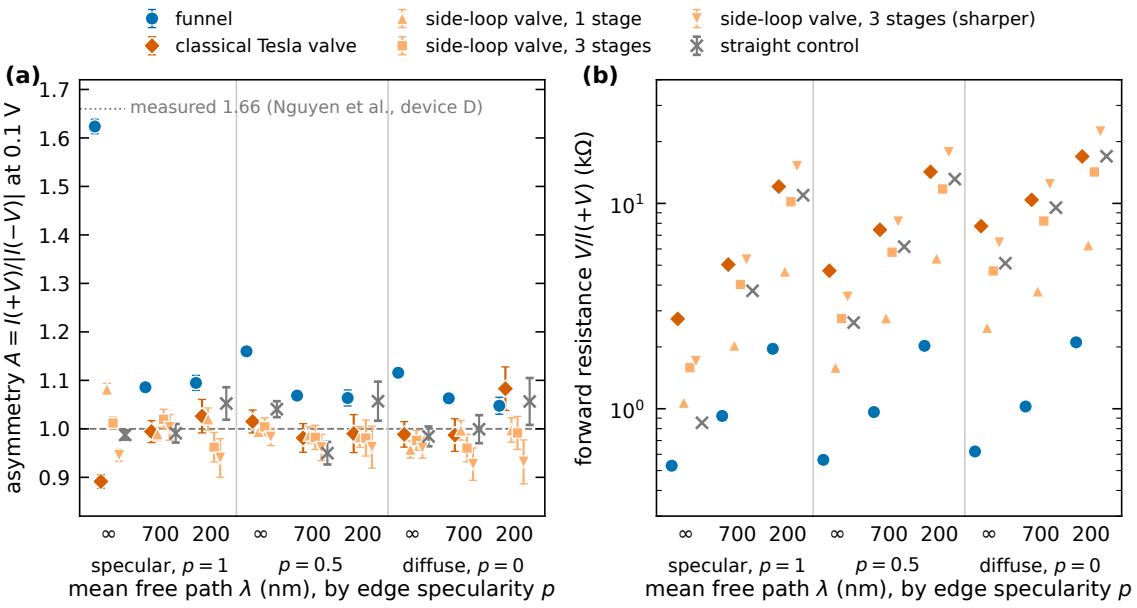
funnel
classical Tesla valve
side-loop valve, 1 stage
side-loop valve, 3 stages
side-loop valve, 3 stages (sharper)
straight control
(a)
measured 1.66 (Nguyen et al., device D)
asymmetry $A = I(+V)/I(-V)$ at 0.1 V
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
∞ 700 200 ∞ 700 200 ∞ 700 200
specular, $p = 1$
$p = 0.5$
diffuse, $p = 0$
mean free path $\lambda$ (nm), by edge specularity $p$
(b)
forward resistance $V/I(+V)$ (kΩ)
$10^1$
$10^0$
∞ 700 200 ∞ 700 200 ∞ 700 200
specular, $p = 1$
$p = 0.5$
diffuse, $p = 0$
mean free path $\lambda$ (nm), by edge specularity $p$